Sentiment Analysis of Tourism Reviews: EDA and Baseline Model

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1 Introduction and Data Description

This project performs sentiment analysis on tourism reviews using the Yelp Academic Dataset, classifying reviews into positive, neutral, and negative categories to provide actionable insights for tourism businesses.

1.1 Dataset Overview

We work with two Yelp Academic datasets: **Business Dataset** (150,346 records, 14 features) and **Reviews Dataset** (7M reviews with text, ratings, and IDs). Our analysis combines structured data (categories, star ratings) with unstructured text data to link sentiment with tourism business characteristics.

2 Data Exploration and Preprocessing

2.1 Methodology

Our EDA followed a three-stage pipeline:

Stage 1: Business Data Cleaning – Removed 12 irrelevant features, dropped records with missing categories, eliminated duplicates by `business_id`, and filtered for tourism keywords (Hotels, Resorts, Attractions, Tours, Museums, Parks, Beaches, Restaurants).

Stage 2: Review Processing – Sampled 7M reviews using chunk processing, retained essential columns (`review_id`, `business_id`, `stars`, `text`), removed duplicates and short reviews (<5 characters).

Stage 3: Integration & Labeling – Merged reviews with tourism businesses via left join, converted stars to sentiment (1-2: negative, 3: neutral, 4-5: positive), removed identifiers post-merge.

2.2 Quality Assurance

Data reconciliation included: duplicate verification (zero post-cleaning), missing value checks (none in critical fields), random shuffling (seed 42), and stratified 80-20 train-test split.

Text preprocessing: lowercase conversion, URL removal, punctuation/number stripping, whitespace normalization, TF-IDF vectorization (10,000 features, bigrams, English stop words removed).

3 Key EDA Findings

Class Distribution: Positive sentiment dominates (typical for tourism platforms), negative is less common (users prefer strong opinions), neutral exists but is minority. Imbalance is addressed via stratified splitting.

Text Characteristics: Reviews range 20-200 words typically. Length distributions vary by sentiment class. Very short reviews (<5 words) were filtered for lacking sentiment information.

Tourism Coverage: Dataset focuses on hospitality (hotels, resorts), dining (restaurants), attractions/landmarks, and travel services, ensuring domain-specific model training.

4 Revised Project Question

Based on EDA insights, our refined question is: *"Can we accurately classify tourism review sentiment (positive, neutral, negative) using review text alone, and what text features most predict customer satisfaction in tourism services?"* This addresses automated classification feasibility, key linguistic indicators, and real-time monitoring potential.

5 Baseline Model

5.1 Architecture & Configuration

We implemented **Logistic Regression** as baseline: Multinomial algorithm, SAGA solver (large dataset optimization), 100 iterations, parallel processing (all cores), 10,000 TF-IDF features.

5.2 Performance

Results on held-out test set:
Model Accuracy: 0.8788583997579605 [87.88%]

Table 1: Baseline Model Performance Metrics

Class	Precision	Recall	F1-Score	Support
Negative	0.84	0.88	0.86	321892
Neutral	0.55	0.31	0.40	138156
Positive	0.92	0.96	0.94	934766
Accuracy		**To be computed**		
Macro Avg	0.77	0.72	0.73	1394814
Weighted Avg	0.86	0.88	0.87	1394814

5.3 Model Interpretation

The logistic regression baseline provides several advantages:

- **Interpretability**: Coefficient weights reveal which terms most strongly predict each sentiment class.
- **Efficiency**: Fast training and prediction times suitable for large datasets.
- **Probabilistic Output**: Provides confidence scores for each prediction via `predict_proba`.
- **Benchmark**: Establishes performance baseline for comparison with more complex models.

5.4 Misclassification Analysis

Initial error analysis indicates:

- Confusion primarily occurs between neutral and positive/negative classes.
- Neutral sentiment is most challenging to classify due to ambiguous language.
- Very short reviews may lack sufficient contextual information for accurate classification.